

Deviation Management

MGS v3.0 | Implementation plan and analysis scope | Derived from the change request workshop deck (24/06/2026) and the synthesis document

1. Purpose in one paragraph

Airbus may authorise a customer to depart from the standard HUMS configuration (raised threshold, changed threshold type). Today this is done by editing the indicator directly, which leaves no trace of the departure. The change replaces direct editing with a first-class **Deviation** object: the departure stays visible, is tied to a TE reference, survives component replacement and relearn, and is reported to Airbus until the configuration is brought back into compliance. Threshold editing through the settings screen disappears entirely – every change now goes through a deviation.

Five deliverables

#	Deliverable	Nature
1	Deviation creation in the enriched Health menu	UI + service + persistence
2	Manage Filtering Laws popup (CRUD, 1–5 contextual parameters)	UI + service + persistence
3	Deviation List tab (active deviations)	UI + query
4	Deviation History tab (deleted deviations)	UI + query + purge
5	DACOM extraction (GSC_DEVIATIONS.sql + THRESHOLDS impact)	SQL + flow

Beyond those five, three cross-cutting behaviours carry most of the technical risk: date-based applicability, the relearn matrix, and acknowledge-on-configuration-update.

2. Implementation steps

Phases are ordered by dependency. Phases 4 and 5 are the ones that touch existing monitoring code and should be prototyped early, not left until the end.

Phase 0 – Analysis and technical design

Walk the existing threshold read path end to end: configuration file import, storage, the settings screen, the learning engine, and session monitoring. Produce the impact list and confirm where a deviation-aware resolution layer can be inserted with a single interception point rather than scattered edits. Close the open points in section 5 with the client before coding.

Phase 1 – Data model and persistence

Create the DEVIATION, FILTERING_LAW and FILTERING_LAW_PARAMETER tables plus the migration script. A deviation carries: aircraft SN, condition indicator reference (with parent 1 / parent 2 path), TE reference, start date, end date (null while active), the deviated threshold fields, and an optional link to a filtering law. Active and historical deviations live in the same table, distinguished by end date – not two tables. Volume is small (~300 rows fleet-wide since 2018), so index for correctness, not throughput.

Phase 2 – Deviation service (create / delete)

Creation: mandatory non-empty TE reference (no format validation), reject a second deviation on an indicator that already has an active one, stamp the start date as now. There is no update operation – the UI must force delete-then-recreate. Deletion sets the end date to now, moves the row to history semantics, and resets the indicator to the configuration defaults.

Phase 3 – Filtering law management

CRUD with a uniqueness check on the law name and one to five contextual parameters. Laws originating from the configuration file, and laws already bound to a deviation, become read-only: shown greyed and non-selectable in the management list, but still available in the indicator dropdown. Delete raises the confirmation popup naming the law.

Phase 4 – Deviation-aware threshold resolution

The highest-risk phase. Resolution must be a function of the session's data date, not the offload date and not the current date: a deviation applies to a session if its start date is before the session date and its end date is null or after it. This means a deleted deviation still applies when replaying an older flight. Implement as a single resolver consulted by monitoring, and cover it with the two worked examples from REQ 12 as unit tests.

Phase 5 – Relearn integration

A deviation set on a learned threshold triggers an automatic relearn request on subsequent flight hours. FIXED to LEARNED and LEARNED to LEARNED with new bounds both relearn, on creation and on deletion alike; LEARNED to FIXED is not applicable. A relearn request must not delete the deviation – verify this explicitly against the existing relearn code.

Phase 6 – UI – Health menu rework

Inline editing of threshold value, minimum threshold and filtering-law parameters directly in the indicator list. Remove the Activate / Remove threshold tick box. Highlight deviated rows in green and block creation on them. Show the default associated law even when no deviation exists. Add the Manage Filtering Laws button and the TE guidance text pointing to AirbusWorld > Request > Technical Request, business domain "DR : Discrepancy Report".

Phase 7 – UI – Deviation List and Deviation History tabs

Both tabs sit alongside the existing Health tab in the MGS settings. List shows active deviations with the full indicator path, only the parameters actually modified, TE reference and application date, plus the delete confirmation popup naming the indicator. History shows deleted ones with the acknowledgment date added, and is empty when nothing deviates or everything is still active.

Phase 8 – Configuration update impact

Applying a new configuration file to an SN acknowledges every deviation on that aircraft. Add the warning popup on the aircraft configuration creation / update screen and on the OK button of the update window, and the per-indicator warning on relearn request creation. Deviations are not migrated – the user recreates them afterwards.

Phase 9 – DACOM extraction

New GSC_DEVIATIONS.sql mirroring the Deviation List for active deviations only: indicator name and path, severity, level, min value, max value, points, filtering law name, filtering law parameters (multi-row), TE reference and creation date. History is never exported. Separately, the existing THRESHOLDS file must carry deviated values transparently with no format change – but must not be regenerated when a deviation touched only a filtering law and not the threshold value itself.

Phase 10 – Purge

Align deviation purge with the existing session purge: delete when the deviation end date is at or before the start date of the oldest non-purged session. Active deviations are never purged.

Phase 11 – Test and validation

Priority on the date-applicability matrix, the relearn matrix in both directions, acknowledge-on-configuration-update, and coverage across both M'ARMS and HELIONIX with EVM indicators (PU and STB) included, since all of those were confirmed in scope during the workshop.

3. Files and classes required for the analysis

No source code was supplied with the two specification documents, so the names below describe **roles** in the MGS code base rather than actual identifiers. Providing these – or simply the package tree around the Health settings and the threshold evaluation – is enough to turn this plan into a concrete, file-by-file impact list.

Priority 1 – blocking, nothing can be scoped without them

Area	What is needed	Why
Condition indicator model	The entity / class representing a condition indicator and its threshold, including the FIXED vs LEARNED distinction and the min / max / points fields	This is the object a deviation overrides. Its shape determines the deviation table columns.
Threshold read path	Every call site that reads a threshold value during session monitoring, and the class that compares indicator values against thresholds	Phase 4 inserts the deviation-aware resolver here. The number of call sites drives the estimate.
Health settings screen	The settings view, its controller, and the table / grid model behind the indicator list	Deliverables 1, 3 and 4 all land in this screen.
Database schema	DDL for the indicator and threshold tables, plus the migration mechanism used between MGS versions	Phase 1 adds three tables and must follow the existing migration convention.

Priority 2 – needed for the impacted behaviours

Area	What is needed	Why
Learning / relearn engine	The relearn request creation path and the class that performs threshold learning over flight hours	Phase 5. Must confirm relearn does not wipe the deviation, and that new bounds are honoured mid-learning.
Configuration import	The configuration file parser and the update-aircraft window with its OK handler	Phase 8 acknowledge-on-update, plus the source of read-only filtering laws.
DACOM extraction layer	The existing GSC_*.sql scripts, the THRESHOLDS file generator and its trigger / scheduler	Phase 9. GSC_DEVIATIONS.sql should follow the existing conventions rather than invent new ones.
Session and purge	The session entity (specifically the data date vs offload date fields) and the existing purge job	Phases 4 and 10 both key off the session data date.

Priority 3 – cross-cutting, needed to match house style

Area	What is needed	Why
Dialogs and i18n	The confirmation-popup utility and the resource bundles	Several new popups with exact wording, in the languages MGS already ships.
Access control	The role / permission model for the settings screens	To confirm who may create and delete a deviation.

4. Proposed data model

Object	Key fields
DEVIATION	id; aircraft SN; condition indicator reference; indicator path (parent 1, parent 2); TE reference; start date; end date (null = active); deviated threshold value; deviated minimum threshold; severity; level; points; filtering law id (nullable)
FILTERING_LAW	id; name (unique); law type (e.g. NR); read-only flag (from configuration file); source (configuration vs user-created)

Object	Key fields
FILTERING_LAW_PARAMETER	id; filtering law id; parameter name; min; max; standard deviation (s). One to five rows per law

A law becomes effectively read-only either because it came from the configuration file or because at least one deviation references it. The second case is a derived condition, not a stored flag.

5. Open points to close before build

- Abdel's GSC_CONFIGURATION.sql proposal (full aircraft configuration extract, including the flight sessions used for learning) is still unestimated. Its field list and extraction frequency need specifying before it can be quoted.
- For a LEARNED threshold whose learning had already completed when the deviation is created, the workshop answer is that a bound change applies even after learning ends – but whether the already-learned value is re-clamped to the new bounds or a fresh learning is launched should be confirmed against the MGS v2.0 correction.
- The wording "session date" was flagged as ambiguous during the workshop and a clearer term was requested; the specification should settle on one before the requirements are frozen.
- The Filter Value column content (maximum number of points, versus the filtering rules with their context parameter, bounds and standard deviation) was to be clarified in the requirement text.

6. Confirmed scope decisions from the workshop

Point	Decision
EVM indicators (PU and STB)	In scope
Aircraft types	Both M'ARMS and HELIONIX
Deviation start date	Immediate only – no past or future date selectable
TE reference format	TE-2026-EC155-00010 style; no validation rule to implement
Threshold editing via settings	No longer possible; everything goes through a deviation
Indicator identification	Name alone is insufficient; the parent component path must be displayed
Deviation values displayed	Only the parameters actually modified
Relearn on learned threshold	Yes, over subsequent flight hours
Volume	~300 deviations fleet-wide since 2018 (3 to 4 per aircraft)